



Economic Analysis Model: User Guide

NOTE: You must have macros enabled in Excel to ensure that all functions of the analysis work correctly. To ensure that macros are enabled in Excel 2010, click the “File” tab, select options from the list on the left side of the screen, select “Trust Center”, then click on “Trust Center Settings”. On the “Trust Center” window, select “Macro Settings”, and then click “Enable all macros”.

Analysis Types

NOTE: The Simple analysis will provide an understandable valuation in most contexts. Only use the other models if you are very comfortable with both DCF calculation and making very quantifiable assumptions about a target firm’s financials.

Simple: Discounted Cash Flow Model of a target company. This model can either be set to weigh future cash flows by probability for discrete outcomes, or to account for continuous risk with a premium in the discount rate. Base year revenue, cost of goods sold, EBITDA and revenue growth are required as inputs.

Complex: Expands on the Simple Analysis Model by including assumption options to calculate terminal value and use different margins in discrete scenarios. A WACC Rate can be calculated for discrete risk in place of a single premium, and discount rates can be set to change over the course of the investment horizon.

Complex/Public Data Model: Offers the same options as the Complex Model, but relies on financial data from Mint Global instead of user-entered information. Up to five financial ratios are flagged on the dashboard, which highlight potential integration problems between the acquirer and the target firm.

Basic Terminology

EBITDA: A non-accounting financial measure, standing for Earnings Before Interest, Tax, Depreciation and Amortization, EBITDA effectively measures operating income while negating aggressive accounting policies. Some investment markets, especially private equity, will value a firm based on a multiple of EBITDA.

Risk-Free Rate: The return an investor can earn by investing in a security with no default risk, typically represented by the 10-year bond rates of the United States Treasury.

Terminal Value: The future value of a firm under the assumption that it will operate with sustainable growth in perpetuity. This value can also be interpreted as a liquidation value, or

This user guide and the corresponding analysis model were developed by Gavin Cochran as part of an undergraduate research project for the University of Washington’s Bothell School of Business, completed during Spring Quarter 2013, under the supervision of Dr. Brandon Fleming.

what the company can be sold for in future dollars. If a firm is expected to have any value at the end of the valuation's time horizon, a terminal value should be calculated.

Time Horizon: The number of periods (months, weeks, etc.) that will be taken into account in a valuation. Will a longer horizon may give a more accurate valuation in regard to financial changes, the horizon should not be set to a point so far in the future that informed estimates cannot be made.

Weighted Average Cost of Capital (WACC): Discount rate of a firm based on current (or expected) capital structure. Weights are preferably calculated based on the current market value of both debt and equity, but book value is acceptable when market information isn't available.

Types of Risk

Continuous: Risk that is ongoing and cannot be quantified as a single event, such as inflation or political risk. Continuous risk is best accounted for by incorporating a premium into the discount rate.

Discrete: Risk that is best quantified by single events, where differing outcomes will result in different values. Examples include regulatory approval for pharmaceutical companies or the passing of specific laws that affect firm prospects. These risks are best measured by calculating a weighted average of the cash flow from each outcome and their probability.

Using the Simple Analysis Model

Selection

From the "Dashboard" tab, click the "Simple Analysis" button in the upper-left corner.

Entering Information

All gray cells require user-input to perform the valuation, all other values are automatically calculated:

	Assumptions
Time Horizon of acquisition/Integration	10
Appropriate analysis of risk	3%
Risk-Free Rate	3%

Provide the base-year revenue, cost of goods sold, EBITDA and historical revenue growth for both the target and the acquirer, which will be the starting point for future projections. "Base year" doesn't necessarily mean the most recent year of available financial data, but one that is best representative of firm performance. Historical revenue growth can be calculated over any time period, but should only include periods that reflect the current state of the firm (calculating revenue growth in periods prior to business divestitures, for example, would not be a good representation of a firm today).

Enter the Time horizon best suited for the valuation, from a range of one period to ten. Only use a time period where you are comfortable with your estimates (If financial estimates can only be made for the next five years, for example, a horizon of ten years would be inappropriate).

Enter the risk-free rate to use in the valuation; remember that while treasury rates are a good proxy for this measure, a historical rate may be more sensible than a current one, depending on current market conditions.

Select whether a discrete or continuous risk type is more suitable in the valuation.

- With a discrete risk type, you will be prompted for the probability of both an optimistic and pessimistic scenario, as well as the expected growth rate of each. Both inputs should be reasonable, and remember that growth can be negative in a pessimistic scenario.

- A Continuous risk scenario will prompt you for a risk premium to add to the discount rate, and assume a status quo growth rate. While there is no singular, preferred way to calculate this value, the risk measure should make sense. Emerging market companies, for example, should have a higher premium than those of developed markets.

After all information is entered, the value of the opportunity is shown in the upper left hand corner. Remember that different assumptions about the target and its riskiness will have a strong impact on this value, so be prepared to defend your thought process!



Additional Tabs for support

The “Combined Company Value” tab can be used to show what portion of company sales would be attributed to the target firm after acquisition at different points in the investment horizon. Analysts can use this tab to show that a target with high growth may make up a large portion of company sales in the near future.

Variables of Combined Company

Revenue	11000.00
Cost of Goods Sold	4167.00
Gross Profit	6833.00
Gross Margin	62.12%
New Revenue from target as a percentage of my company - 1 year	22.22%
New Revenue from target as a percentage of my company - 1 year	261.39%
Growth prospects of combined company (accounting for percentage)	
Expected	
Sales Growth - 1 year	6.76%
Sales Growth - 10 years	20.67%
Sales growth over 10 years	246.54%

The “Variables of Acquirer” and “Variables of Target” tab can be used to summarize the current state of each company. Further, the visuals on each tab can be used in a discrete risk analysis to show what sales and profits will look like in different scenarios.

Using the Complex Analysis Model

Selection

From the “Dashboard” tab, click the “Complex Analysis” button in the upper-left corner.

Entering Information

All gray cells require user-input to perform the valuation, all other values are automatically calculated:

Time Horizon of acquisition/Integration	Assumptions	10
Appropriate analysis of risk		3%
Risk-Free Rate		

Provide the base-year revenue, cost of goods sold, EBITDA and historical revenue growth for both the target and the acquirer, which will be the starting point for future projections. “Base year” doesn’t necessarily mean the most recent year of available financial data, but one that is best representative of firm performance. Historical revenue growth can be calculated over any time period, but should only include periods that reflect the current state of the firm (calculating revenue growth in periods prior to business divestitures, for example, would not be a good representation of a firm today).

Enter the Time horizon best suited for the valuation, from a range of one period to ten. Only use a time period where you are comfortable with your estimates (If financial estimates can only be made for the next five years, for example, a horizon of ten years would be inappropriate).

Decide whether or not the target will have a terminal (or liquidation) value at the end of the investment horizon, which is calculated with following equation:

$$\text{Terminal Value} = \frac{CF_n(1 + g)}{(WACC - g)}$$

Where n is the final year of the investment horizon, and g is the long-term growth rate assuming the firm will exist forever. Remember that while Terminal value is calculated with the cash flow from the year following the investment horizon, the value is actually discounted back one year and can be combined with the previous year's cash flow.

Enter the sustainable revenue growth rate of the firm; remember that firm history is not necessarily indicative of a sustainable growth rate, and economy-wide measures may be more appropriate in your assumptions. A firm, for example, can never grow faster than the economy in perpetuity, and as-such the project GDP growth rate of the target firm's economy may serve as a good cap to this rate. In more mature targets, multiplying the firm's return on invested capital (ROIC) by the retention ratio (how much the firm retains to grow) may be a good measure of long-term growth.

Enter the risk-free rate to use in the valuation; remember that while treasury rates are a good proxy for this measure, a historical rate may be more sensible than a current one, depending on current market conditions.

Select whether a discrete or continuous risk type is more suitable in the valuation.

-With a discrete risk type, you will be prompted for the probability of both an optimistic and pessimistic scenario, as well as the expected growth rate of each. Both inputs should be reasonable, and remember that growth can be negative in a pessimistic scenario. You will also be asked to decide whether margins (gross and EBITDA) are expected to change over time. If they are, enter optimistic and pessimistic estimates for both the target and the acquirer.

-A Continuous risk scenario will prompt you to calculate the weighted average cost of capital for the target firm, based on the following formula:

$$WACC = \frac{E}{V} \text{Cost of Equity} + \frac{D}{V} \text{Cost of Debt}(1 - \text{Marginal Tax Rate})$$

Enter in as much information about the capital structure of the firm as possible; market rates are preferable in calculation ratios, but book values are an acceptable alternative. The Equity risk premium can be found with the following equation:

$$\text{Equity Risk Premium} = \beta(\text{Market Return} - R_f)$$

Where R_f is the risk-free rate, and β is the Beta coefficient of the target's stock performance against that of the market. Market return can be calculated by tracking changes on the S&P 500 over a given time-period and computing an average return, though other indexes may be more appropriate for certain firms. Remember that the historical data used in such measures must be applicable to the market as it is today (S&P

500 returns from the 1930's, for example, would not be appropriate in calculating market return). Combining the equity risk premium with the risk-free rate provides a firm's cost of equity.

The cost of debt is calculated by attaching a credit default spread to the risk-free rate in the market; the quoted returns on any of the firm's outstanding bonds can give an idea of what this spread may be, but the ratings criteria of agencies such as Moody's and Standard & Poor can be used to find an appropriate credit spread. With this method, compare the target's financial ratios to those listed by the ratings agency to identify a credit rating. Each credit rating comes with a different default spread.

Enter the marginal tax rate in order to calculate the after-tax cost of debt (since debt is a tax deduction); this value changes rarely, and has historically sat at 35%.

Finally, select whether the target's risk will remain the same through the life of the investment, or change overtime. If the firm's capital structure or market risk preferences are expected to change, the risk level of the firm will most likely change over time. Fill in the expected discount rate for each future period; if you want to test these assumptions over a different investment horizon, you'll need to change the number of periods and then select the assumptions again.

HINT: You can recalculate a new cost-of-capital for use in the varying risk assumption with the steps outlined above; after entering in each value for the WACC, go to the "Combined Company Value-Complex" tab and copy the value from range B22. You can enter this value in the matrix of risk rates. If a rate in any future period will be the same as the one used in initially, copy the resulting value from range B22 anyway, as some value must be entered for each period.

After all information is entered, the value of the opportunity is shown in the upper left hand corner. Remember that different assumptions about the target and its riskiness will have a strong impact on this value, so be prepared to defend your thought process!



Additional Tabs for support

The "Combined Company Value" tab can be used to show what portion of company sales would be attributed to the target firm after acquisition at different points in the investment horizon.

Analysts can use this tab to show that a target with high growth may make up a large portion of company sales in the near future.

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The “Variables of Acquirer” and “Variables of Target” tab can be used to summarize the current state of each company. Further, the visuals on each tab can be used in a discrete risk analysis to show what sales and profits will look like over time in different scenarios.

Using the Public Data Model

Selection

From the “Dashboard” tab, click the “Public Data” button in the upper-left corner.

Pulling Public Data from Mint

From the “UWB Libraries” Homepage, search for “Mint Global”

UW Libraries · UW Bothell · UW Seattle · Cascadia · MyUW

CAMPUSLIBRARY

Serving University of Washington Bothell and Cascadia Community College

RESOURCES SERVICES ABOUT THE LIBRARY YOUR LIBRARY ACCOUNT ASK US A QUESTION

off-campus access (logged in)

The Campus Library will be closed Monday May 27th for Memorial Day and will close at 5:00pm Sunday May 26th.

RESOURCES

- » Research Guides by Subject
- » Class Guides: CCC, UWB
- » Reference Tools
- » Databases A-Z: articles & more
- » Study Room Reservations
- » Reserves/Course Readings
- » More resources...

UW WorldCat: Search UW Libraries and beyond

Books Video Audio Articles UWB-CCC All

Mint Global Search All

Advanced search | Help

Search with WorldCat

UW-only Catalog

SERVICES

- » Ask a Question
- » Connect from Off-Campus
- » Interlibrary Loan: CCC, UWB
- » Borrowing Information
- » Faculty Guide
- » Student Guide
- » More services...

ABOUT THE LIBRARY

- » Hours and Contact Info
- » Maps, Directions, and Bus info
- » Policies
- » Subject Librarians
- » Library Art Tour
- » New Books & Media
- » More about the library...

UW Bothell UNDERGRADUATES NEEDED!

Services, Resources, Spaces

Take the 2013 Survey of Student Library & Information Use

<http://tinyurl.com/uwblibrarysurvey>

2) After being redirected to the UW Library site, select the first result by Bureau Van Dijk Electronic Publishing

Search results for "Mint Global" linked to Libraries Worldwide

Results 1-10 of about 888 (67 records)

Sort by: Library and Services Save Search

Mint Global
by Bureau van Dijk Electronic Publishing
Online · Updating database · Computer File · Continually Updated Resource
Language: English
Publisher: Amsterdam: Bureau van Dijk Electronic Pub.
Database: WorldCat
License: Not over the Internet: University of Washington Libraries

Global focus: U.S. foreign policy at the turn of the millennium
by Martha Honey, Tom Barry
Book
Language: English
Publisher: New York: St. Martin's Press, 2000
Database: WorldCat
License: Not over the Internet: University of Washington Libraries
Editions and formats: 1

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 **MARKET RESEARCH**
Industry profiles from Datamonitor

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Images from Global Reports, EOL and original documents

MintGlobal

[Add Mint Global in search bar](#)

 BUREAU VAN DIJK

Search for your desired company and click on the appropriate result

Run issue: Multiple results for same company name may appear, depending on the firm's organizational structure. The Country Code, last available year of data and revenue figure can assist students in selecting the proper result

From the “Build a Report” page, scroll to the bottom of the list, clear all selections, check “Financials-Detailed Standard Format” and “Cash Flow Statement”, then hit “View”.

MintGlobal / Powerful business intelligence

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You currently have no saved reports. Click on [Save my report](#) to save your current report.

BUILD A REPORT [Save my report](#) [Standard report](#) [Clear all](#) [Cancel](#) [View](#)

☐ Check the boxes by each section in the order you'd like them to be displayed in your report

- ☐ Mergers & acquisitions
- ☐ News
- ☐ Original documents
- ☐ Notes

For most quoted (and some delisted) companies:

- ☒ Financials - Detailed standard format
- ☒ Cash Flow statement
- ☐ Accounting practices
- ☐ Segment data
- ☐ Stock data
 - ☐ Graph - Stock data
- ☐ Financial forecasts & brokers current recommendation
- ☐ Graphs - Financial forecasts
- ☐ Detailed estimates
- ☐ Graphs - Detailed estimates

The following three sections are not available for all quoted companies.

- ☐ Future outlook
- ☐ Major competitors
- ☐ Major customers

☐ This could be your "custom-made" data section in your Mint report.
[Contact us](#) to find out how we can integrate internal and external data sources into Mint.

☐ Bypass this window prior to report display [Save my report](#) [Cancel](#) [View](#)

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On the report output, select “Export” and save as .xls

On the exported files, copy over

Entering Information

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Assumptions

Time Horizon of acquisition/Integration: 10

Appropriate analysis of risk

Risk-Free Rate: 3%

Provide the historical revenue growth for both the target and the acquirer, which will be the starting point for future projections. The Target and Acquirer variables tabs contain various measures of growth based on the historical data provided. You are free to use any method you desire in inputting a growth rate, but you must be able to justify your growth assumption.

Enter the Time horizon best suited for the valuation, from a range of one period to ten. Only use a time period where you are comfortable with your estimates (If financial estimates can only be made for the next five years, for example, a horizon of ten years would be inappropriate).

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Using the Ratio Flag Tool

The Public Data Model is designed to analyze the balance sheets and income statements of both the target and acquiring companies in order to identify concerning areas to take into account during the valuation.

After uploading new data for either firm from Mint, return to the dashboard and set a gap threshold in range G4 as a % (a 15% threshold, for example, would be entered as .15). After setting a threshold, hit the “Update Ratios” tab to re-run a comparison of the firms’ financials, which looks at the following types of data:

- Balance Sheet Changes:** On a percentage basis, how much have specific balance sheet items changed over the course of the historical period?
- Current Balance Sheet:** Based on the most-recent period, how much liquidity and leverage does each firm carry?
- Efficiency:** What returns are the firms earning from their asset and funding bases?
- Capital Utilization:** How well does each firm manage their working capital?

The top five ratios with the greatest gap between the firms, regardless of category, are flagged in cells E7 through E11. The corresponding Pivot Table in cell I6 shows how many items in each data type were flagged based on the percentage threshold entered. Based on the number of items flagged in each bucket, users can determine what differences between the two firms are overlooked in the basic model and should be accounted for. To get a detailed look at the flagged ratios, double-click on the number of flagged values displayed in column J. A new sheet will populate showing all of the ratios flagged for the specific bucket selected. Each ratio can be expanded on in an explanation of the resulting valuation.

Additional Tabs for support

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